Manual

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Contents

1 Flatcam:

1.1 calculate tool diameter

tools→calculator input tip-Diameter input tip angel input desired cut-z use the tool diameter given

1.2 Isolation Routing

File→Open Gerber select the gerber file that has your isolation routing

enter the tool diameter that you obtain from the calculator under isolation routing

select a feed rate between 0 and 170 mm per minute higer feed rate implies faster routing time but rougher cuts (recommended: 170) set a Cut Z average standard copper thickness is 0.05 mm but consult the laminates info set travel Z to 2.00 mm

click generate geometry button

select the generated geometry

click on the Export button under "export G-code"

1.3 Drilling

File→Open Excellon

select the tools you want to create the CNC job fore

Adjust Drill Z to the thickness of the copper laminate

set travel Z to 2.00 mm

set feed rate between 0 and 170

click Create CNC Job

click on the Export button under "export G-code"

1.4 Board Cutout

File→Open Gerber for your board cutout select your board cutout specify a Margin. This will create a rectangular cutout at the given distance from any element in the Gerber. Specify a Gap Size. 2 times the diameter of the tool you will use for cutting is a good size. Specify how many and where you want the Gaps along the edge, 2 (top and bottom), 2 (left and right) or 4, one on each side. Click on Generate Geometry.

Create a CNC job for the newly created geometry

1.5 2-side PCB

open File→Open Gerber for both top and bottom layers of the PCB

select Tools→Double-sided PCB tool select which mirror axis create a mirror image of the bottom layer by clicking on Create Mirror

create the alignment hole drill object by clicking Create alignment drill in the Double-sided PCB tool

click on the Export button under "export G-code"

select the gerber file that has your bottom layer isolation routing

enter the tool diameter that you obtain from the calculator under isolation routing

select a feed rate between 0 and 170 mm per minute higer feed rate implies faster routing time but rougher cuts (recommended: 170) set a Cut Z average standard copper thickness is 0.05 mm but consult the laminates info set travel Z to 2.00 mm

click generate geometry button

select the generated geometry

click on the Export button under "export G-code"

2 Wegster operation:

2.1 start up

load the g-code files that you wish to use onto the computer in a way that you can find them. Open Wegstr CNC controlling software if it is not already open. if the indicator under status does not read device ready push the black button next to the power button on the wegstr CNCMilling Machine clamp down the copper laminate to the machine bed locate the bits with corresponding dimensions as the tools used in FlatCAM

2.2 safety

The manufacturer recommends the use of eye protection and protective gloves while operating the machine.

2.3 Isolation Routing

change the bit to the spital bit load the g-code for your isolation routing push the button in the software under the text auto level than press the button labeled start use the vacuum to remove the fiberglass recommended to use while the Routing is tacking place

2.4 Drilling

load the g-code for the drilling

change to the drill bit

if the isolation routing has bin used, use the x and y movement controls to move the spindle to x = 0 and y = 0 and than use the z axis control to move the drill bit down to the copper and set z = 0

else push the auto level button

than press the button labeled start use the vacuum to remove the fiberglass recommended to use while the drilling is tacking place

2.5 Board Cutout

this is the last operation for the PCB change the bit to the Flat end milling bit load the g-code for the board cutout move the spindle to 0,0,0 press start use the vacuum to remove the fiberglass recommended to use while the milling is tacking place

2.6 2-side PCB

load the alignment drill g-code change the bit to your drill bit use the x and y movement controls to move the spindle to x = 0 and y = 0 and then use the z axis control to move the drill bit down to the copper and set z = 0 press the start button

declamp the laminate flip the laminate to the other side use the alignment holes to align the laminate clamp the laminate down to the work bed load the bottom isolation routing g-code change to the spiral bit move the spindle to 0,0,0 and press auto level press start button

3 Links:

3.1 flatcam:

http://flatcam.org/manual/index.html

3.2 wegstr manual pdf:

https://testglassware.com/wp-content/uploads/2022/05/MANUAL-ITEM-1.pdf